

SHIVANG SINHA

AI Full Stack Engineer

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SUMMARY

AI Full Stack Engineer with 5+ years delivering production-grade applications end-to-end — Python/FastAPI backends, React/TypeScript frontends, and REST API architectures in enterprise-regulated environments. Currently building an LLM-powered Copilot at Carl Zeiss on Azure; previously shipped high-reliability transactional systems at Barclays (99.9% uptime) and industrial platforms at Hexagon (500+ users). Combines deep AI/ML expertise (RAG, multi-agent systems, LLM orchestration) with strong DevSecOps practices across Docker, CI/CD, and Azure. Proven track record leading code reviews, collaborating with vendor teams, and driving AI-assisted development adoption. M.Sc. in NLP (University of Trier, thesis completed).

WORK EXPERIENCE

Generative AI Engineer (Werkstudent)

Nov 2024 – Present

Carl Zeiss Microscopy

Munich, Germany

- Designed and deployed an LLM-driven Microscopy Copilot on Azure OpenAI with LangGraph, architecting both the FastAPI backend and React frontend, reducing manual R&D workflows by 25–35%
- Built RAG pipelines with vector databases for intelligent retrieval across documentation and code repositories, improving query accuracy by ~40%
- Developed multi-agent systems for NL-to-code translation, automating Python and C# script generation with structured validation and integration contracts
- Shipped production features via MCP-based microservices with CI/CD pipelines and containerized deployments, driving cross-department adoption
- Conducted code reviews and design sessions with cross-functional teams, ensuring quality, maintainability, and alignment with architectural standards

Senior Software Developer

Mar 2022 – Mar 2024

Barclays

Pune, India

- Led full-stack development of enterprise web applications for investment banking, ensuring scalability and compliance with regulatory standards
- Designed and delivered RESTful APIs for high-volume transactional workflows, governing integration contracts between systems and achieving 99.9% production reliability
- Co-built an NLP chatbot with Python backend and API integrations, automating FAQ responses and reducing support tickets by 30%
- Drove structured code reviews, sprint planning, and secure release processes in Agile Scrum teams of 6–8 engineers, mentoring junior developers
- Collaborated with external vendor teams, reviewing deliverables and holding implementations to agreed quality standards

Software Developer

Sep 2020 – Mar 2022

Hexagon CCI

Hyderabad, India

- Launched customer-facing industrial web applications (frontend + backend) for manufacturing clients, serving 500+ enterprise users
- Optimized backend services and REST APIs powering distributed systems, improving response times by 20% through query optimization and caching

EDUCATION

M.Sc. Natural Language Processing

Apr 2024 – Mar 2026

University of Trier

Trier, Germany

Thesis: Comparative Study of NLP Models for Analysis of AI/ML Skill Demand in the German Job Market

B.E. Computer Science

2016 – 2020

Chandigarh University

Chandigarh, India

AVAILABILITY

- Valid German residence permit; open to EU relocation
- Full-time from March 2026

TECHNICAL SKILLS

Backend Development

Python, FastAPI, Flask, Django, Pydantic, Celery, REST API design, async processing

Frontend Development

React, TypeScript, JavaScript, SPA architecture, component libraries, API integration patterns

AI / Machine Learning

LLMs, RAG, AI Agents, Agentic Workflows, Prompt Engineering, LLM Orchestration, NLP, Fine-tuning

Frameworks & Tools

LangChain, LangGraph, HuggingFace, PyTorch, spaCy, BERT, MCP, Vector Embeddings

Cloud & DevSecOps

Microsoft Azure, Azure OpenAI, Docker, CI/CD pipelines, containerized deployments, MLOps

Databases

SQL, PostgreSQL, MongoDB, Vector DBs (FAISS, Chroma)

AI-Assisted Dev Tools

GitHub Copilot, Cursor, Claude Code

PROJECTS

Agent-Based Research Assistant (2025)

Full-stack multi-agent system: FastAPI backend + React frontend with RAG and real-time source citations.

LLM Document QA System (2024)

Document QA with FAISS vector indexing, REST API layer, and generative LMs for grounded answers.

CERTIFICATIONS

- LangChain Academy – Agentic AI (2026)
- HuggingFace AI Agents Course (2026)
- Microsoft AI Tour – GenAI & Azure (2025)

LANGUAGES

- English** – Fluent (C1)
- German** – Intermediate (B1/B2)